Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/02/2016
PI Name:	Asaithamby, Aroumougame Ph.D.		
Project Title:	Mechanism of Clustered DNA Double-strand Break Repair in Response to HZE Particles in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7201	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2017 per S. Monk/LaRC (Ed., 11/7/16) NOTE: Extended to 12/31/2016 per NSSC information (Ed., 10/5/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chen, David (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX13AD57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	DNA damage and consequent mutations initiate the multi-step carcinogenic process. Substantial evidence indicates that high charge and energy [HZE] particles, component of galactic cosmic rays (GCR), induced clustered DNA damages are difficult to repair, and in some instances are irreparable; this has been associated with the increased relative biological effectiveness of chromosomal aberrations, mutagenesis, and carcinogenesis. However, it is unclear why clustered lesions are difficult to repair, and the mechanisms by which unrepaired clustered DNA lesions trigger carcinogenic events are not known. The goals of this proposed research are to utilize our unique capability to quantitatively estimate the ability of cells to repair clustered DNA lesions in response to a spectrum of HZE particles, and to dissect the genetic factors that may potentially be involved in the processing of complex DNA lesions. The specific aims of the proposed research are to test the hypothesis that: (1) the cellular ability to repair HZE particles induced clustered DNA damage is modulated by radiation quality and cell cycle stages, utilizing our newly established in situ and imaging technology; (2) homologous recombination plays a key function in facilitating the repair of double-strand breaks associated with clustered DNA lesions, using key homologous recombination factors deficient cells; (3) Fanconi Anemia (FA) complex may orchestrate the repair of complex double strand breaks (DSBs) induced by HZE particles, using a panel of FA pathway deficient human cells; and (4) DNA end-processing nucleases, such as Werner Syndrome protein (WRN) and Artemis, may facilitate the processing of complex DNA ends generated by HZE particles using human cells that are either deficient in WRN and Artemis or expressing different nuclease mutant forms of WRN and Artemis. Understanding the mechanism underlying the repair of DNA damage induced by HZE particles will have important implications for estimating the risks to Astronaut health associated with GCR exposures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the mechanism underlying the repair of DNA lesions induced by HZE particles will have important implications for estimating the risks to Astronaut's health associated with GCR exposures. Our research would also lead to the development of counter-measures to enhance astronaut’s ability to sustain HZE particles exposures in space.
Task Progress:	1. We identify a novel role for Rad51 in acute immune signaling in response to space radiation. 2. Recruitment of Rad51 to perturbed replications forks prevents MRE11-mediated degradation of nascent DNA strands and recruitment to DSBs facilitates the repair of DSBs through a HR-mediated repair pathway. 3. In the absence of Rad51, fragmented newly replicated genomic DNA, and possibly together with defective DSBs repair, is released into cytosol triggering STING-mediated acute immune response. 4. The coordinated activities of Rad51 in DNA repair and replication fork processing and inhibition of immune signaling are critical for the suppression of chromosome instability. 5. These results can be exploited to mitigate space radiation induced carcinogenesis.
Bibliography Type:	Description: (Last Updated: 03/12/2018)
Abstracts for Journals and Proceedings	Asaithamby A. "Rad51 interconnects between DNA replication, DNA repair and immunity." 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. , Oct-2016
Articles in Peer-reviewed Journals	McFadden CH, Hallacy TM, Flint DB, Granville DA, Asaithamby A, Sahoo N, Akselrod MS, Sawakuchi GO. "Time-lapse monitoring of DNA damage colocalized with particle tracks in single living cells." Int J Radiat Oncol Biol Phys. 2016 Sep 1;96(1):221-7. Epub 2016 Apr 16. http://dx.doi.org/10.1016/j.ijrobp.2016.04.007 ; PubMed PMID: 27511858 , Sep-2016
Articles in Peer-reviewed Journals	Su F, Bhattacharya S, Abdisalaam S, Mukherjee S, Yajima H, Yang Y, Mishra R, Srinivasan K, Ghose S, Chen DJ, Yannone SM, Asaithamby A. "Replication stress induced site-specific phosphorylation targets WRN to the ubiquitin-proteasome pathway." Oncotarget. 2016 Jan 5;7(1):46-65. http://dx.doi.org/10.18632/oncotarget.6659 ; PubMed PMID: 26695548; PubMed Central PMCID: PMC4807982 , Jan-2016
Articles in Peer-reviewed Journals	Sridharan DM, Asaithamby A, Blattnig SR, Costes SV, Doetsch PW, Dynan WS, Hahnfeldt P, Hlatky L, Kidane Y, Kronenberg A, Naidu MD, Peterson LE, Plante I, Ponomarev AL, Saha J, Snijders AM, Srinivasan K, Tang J, Werner E, Pluth JM. "Evaluating biomarkers to model cancer risk post cosmic ray exposure." Life Sci Space Res (Amst). 2016 Jun;9:19-47. http://dx.doi.org/10.1016/j.lssr.2016.05.004 ; PubMed PMID: 27345199 , Jun-2016
Articles in Peer-reviewed Journals	Bhattacharya S, Asaithamby A. "Ionizing radiation and heart risks." Semin Cell Dev Biol. 2016 Oct;58:14-25. Epub 2016 Feb 2. Review. http://dx.doi.org/10.1016/j.semcdb.2016.01.045 ; PubMed PMID: 26849909 , Oct-2016